NLP WORKSHOP SERIES, AN LLM INSIGHT

LLaMA

IN THIS WORKSHOP

- Provide a guided tour of Llama:
  - Introduction to Large Language Models
  - What is Llama2?
  - How to access Llama2
  - Prompt Engineering
  - RAG (Retrieval Augmented Generation)
  - Fine-Tuning

LARGE LANGUAGE MODELS (LLMs)

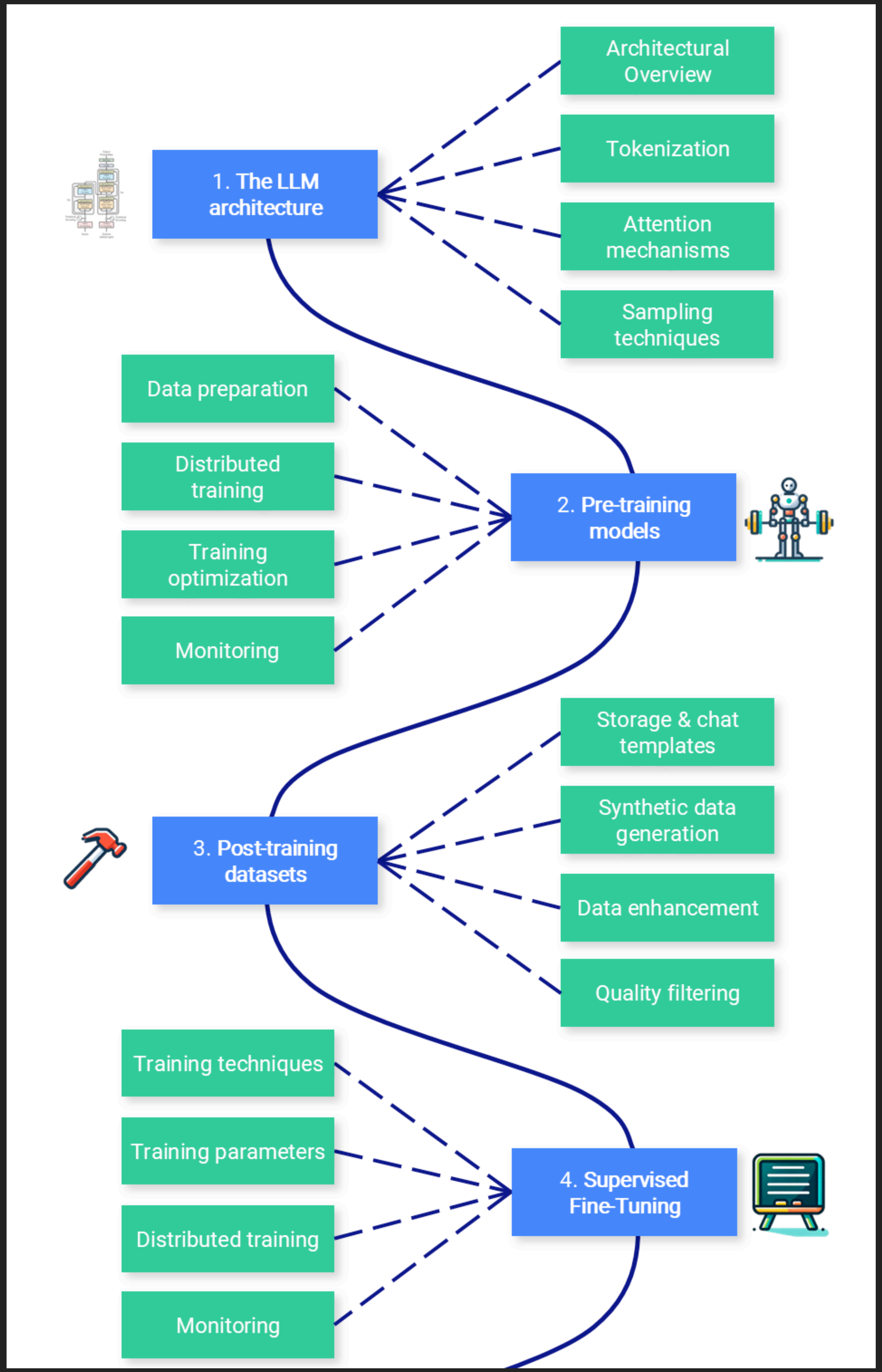
- Large language models are sophisticated artificial intelligence systems designed to understand and generate human-like text.
- They are built on deep learning architectures, enabling them to process and generate human-like text based on patterns in data.
- Characteristics:

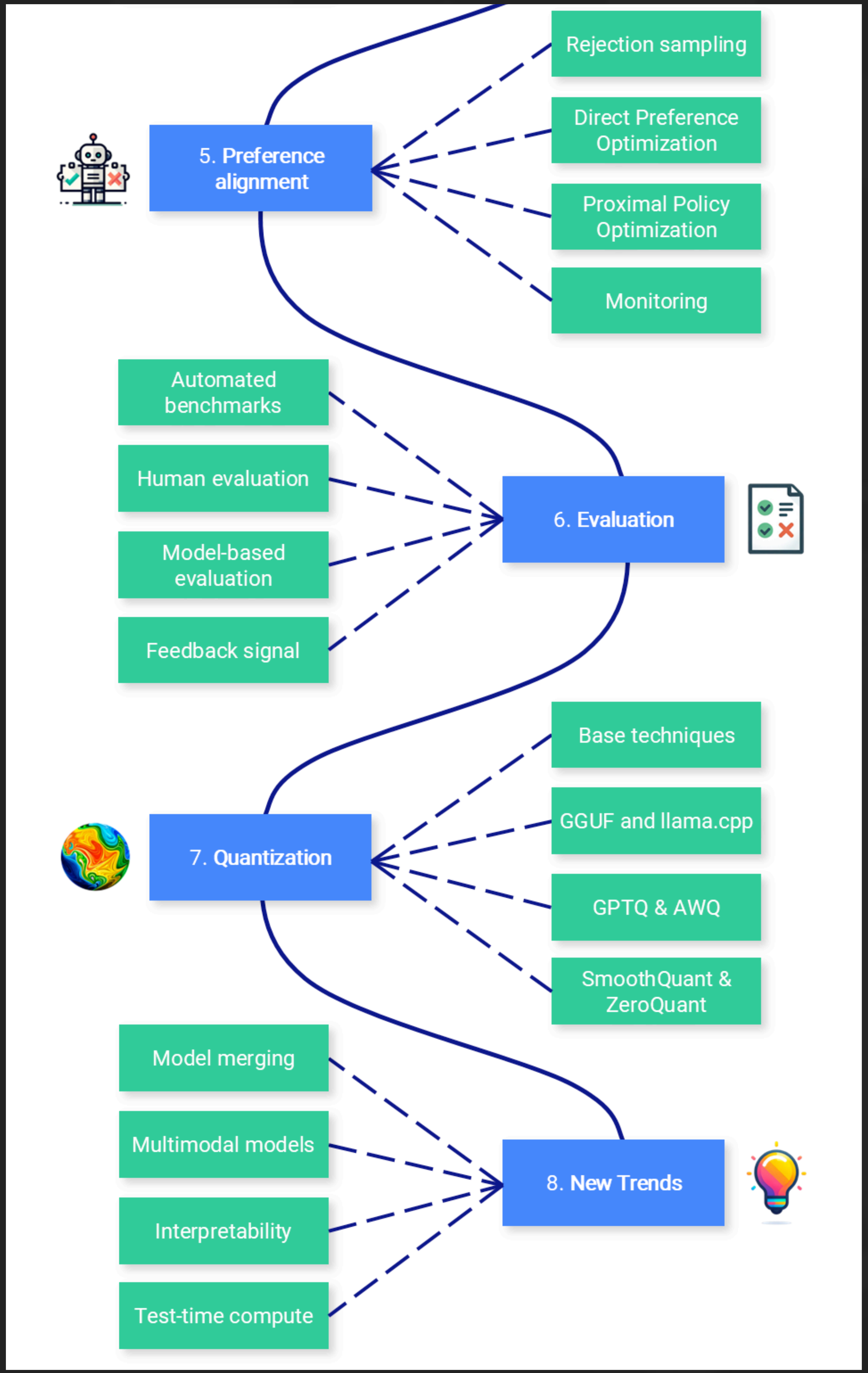
MASSIVE DATA TRAINING

COMPLEX ARCHITECTURES

NATURAL LANGUAGE UNDERSTANDING

ROADMAP


1. The LLM architecture
Architectural Overview
Tokenization
Attention mechanisms
Sampling techniques
2. Pre-training models
Data preparation
Distributed training
Training optimization
Monitoring
3. Post-training datasets
Storage & chat templates
Synthetic data generation
Data enhancement
Quality filtering
4. Supervised Fine-Tuning
Training techniques
Training parameters
Distributed training
Monitoring


5. Preference alignment
Rejection sampling
Direct Preference Optimization
Proximal Policy Optimization
Monitoring
6. Evaluation
Automated benchmarks
Human evaluation
Model-based evaluation
Feedback signal
7. Quantization
Base techniques
GGUF and llama.cpp
GPTQ & AWQ
SmoothQuant & ZeroQuant
8. New Trends
Model merging
Multimodal models
Interpretability
Test-time compute

LLM VISUALIZATION

- What is happening in the background?
- https://bbycroft.net/llm

WHAT IS LLAMA2 ?

- Large Language Model Meta AI
- Next Generation of Llama (7 Feb 2023)
- SOT of Open Source LLMs provided by MetaAI (publicly available datasets)
- 7B, 13B, 70B: Accuracy ~ Cost
- Pre-trained and Chat (Optimized for dialog use cases)
- Research Paper (19 July 2023 ~5 months later)

Llama 2: Open Foundation and Fine-Tuned Chat Models

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Peter Albert Amjad Almahairi Yasmine Babaei Nikolay Bashlykov Soumya Batra
Prajjwal Bhargava Shruti Bhosale Dan Bikel Lukas Blecher Cristian Canton Ferrer Moya Chen

HOW TO ACCESS TO LLAMA2 ?

- Register + download from Meta + self-host

- Hosted API Platform (e.g. Replicate)

- Hosted Container Platform (e.g. Azure, AWS, GCP)

USE CASES

CONTENT GENERATION

PROGRAMMING

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CHATBOTS

SUMMARIZATION

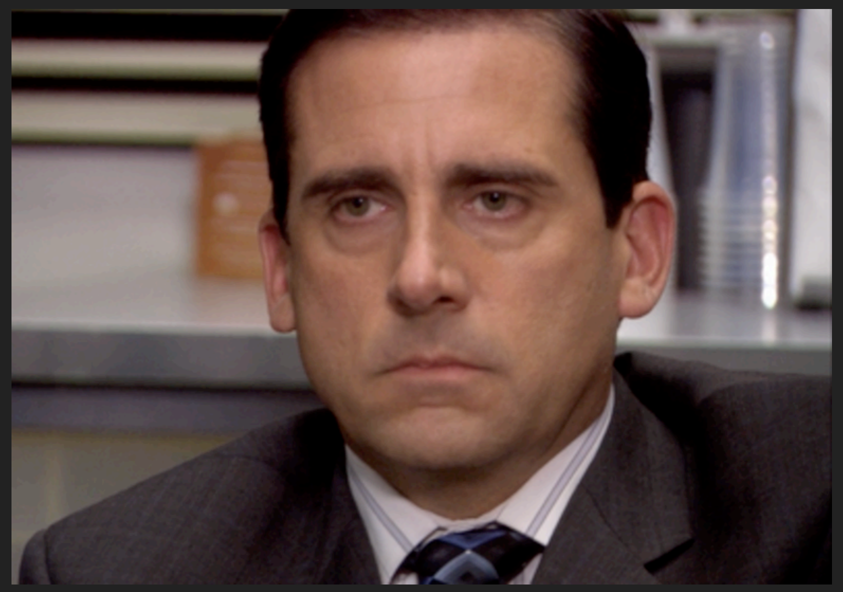
LLMs ARE STATELESS

Hey Llama, I saw Jim today.

Great.

Is he married to Pam?

Who is married to Pam?



PROMPT ENGINEERING

- Refers to the science of designing effective prompts to get the desired response (2048 tokens)

1. In-Context Learning
2. Chain-of-Thought (COT)

PROMPT ENGINEERING:

1) IN-CONTEXT LEARNING

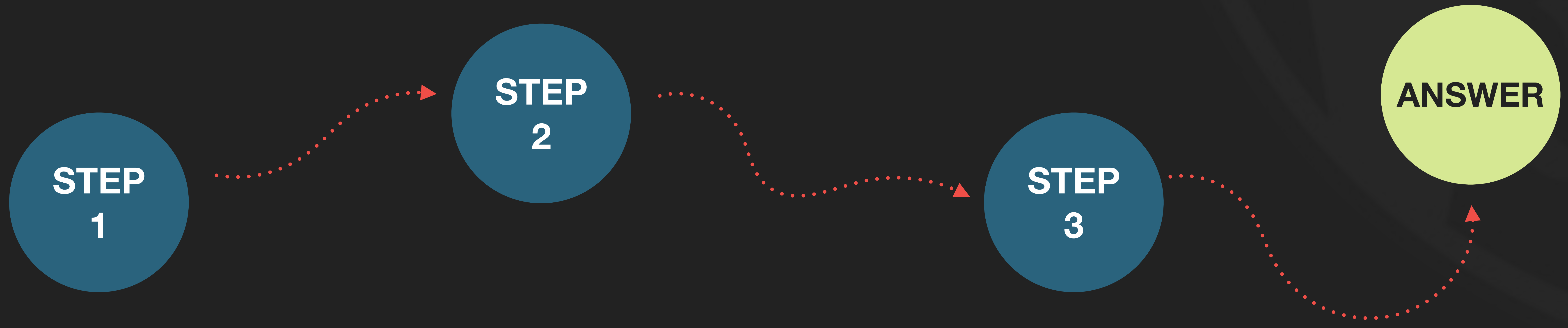
- Specific method of prompt engineering where demonstration of tasks are provided as part of prompt

    1. Zero-Shot learning: model is performing task without any input examples
    2. Few-Shot or N-Shot learning: Model is behaving and performing based on input examples in user's prompt

PROMPT ENGINEERING:

2) CHAIN-OF-THOUGHT

- ‘chain of thought’ or a coherent sequence of ideas is crucial for generating meaningful and contextually relevant responses.
- What was the logical path to get this answer?



PROMPT ENGINEERING LIMITATIONS

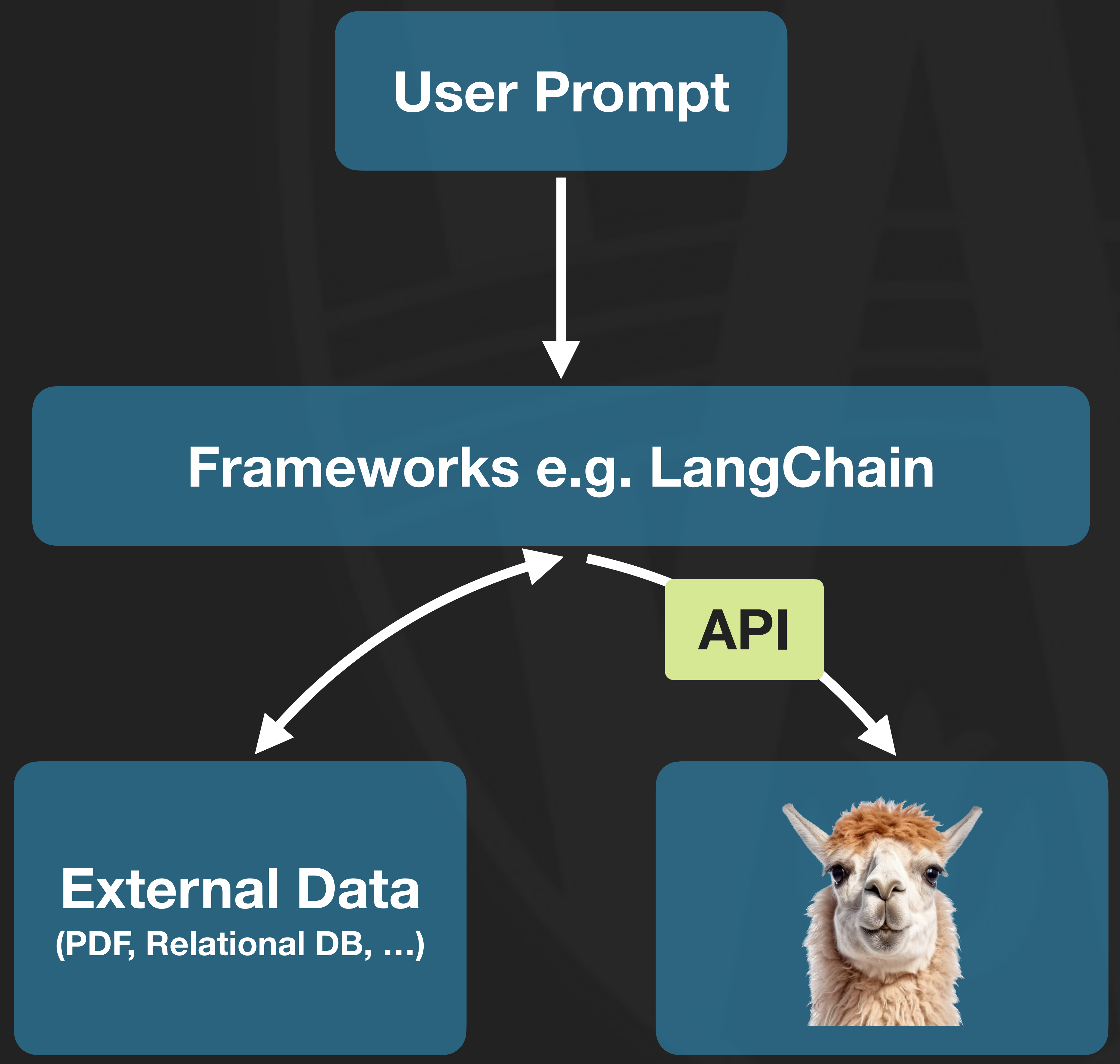
1. Pre-training date
   - LLMs trained up to a certain date. Beyond that date, they are not able to get any inferencing. Think about new records, stats, papers …
2. Lacks of Specialized Knowledge
   - If you have a dataset that is not publicly available, then Llama does not know about that. Sometimes datasets are way more than 2048 tokens and you are not able to use prompt engineering.

- Solution? RAG

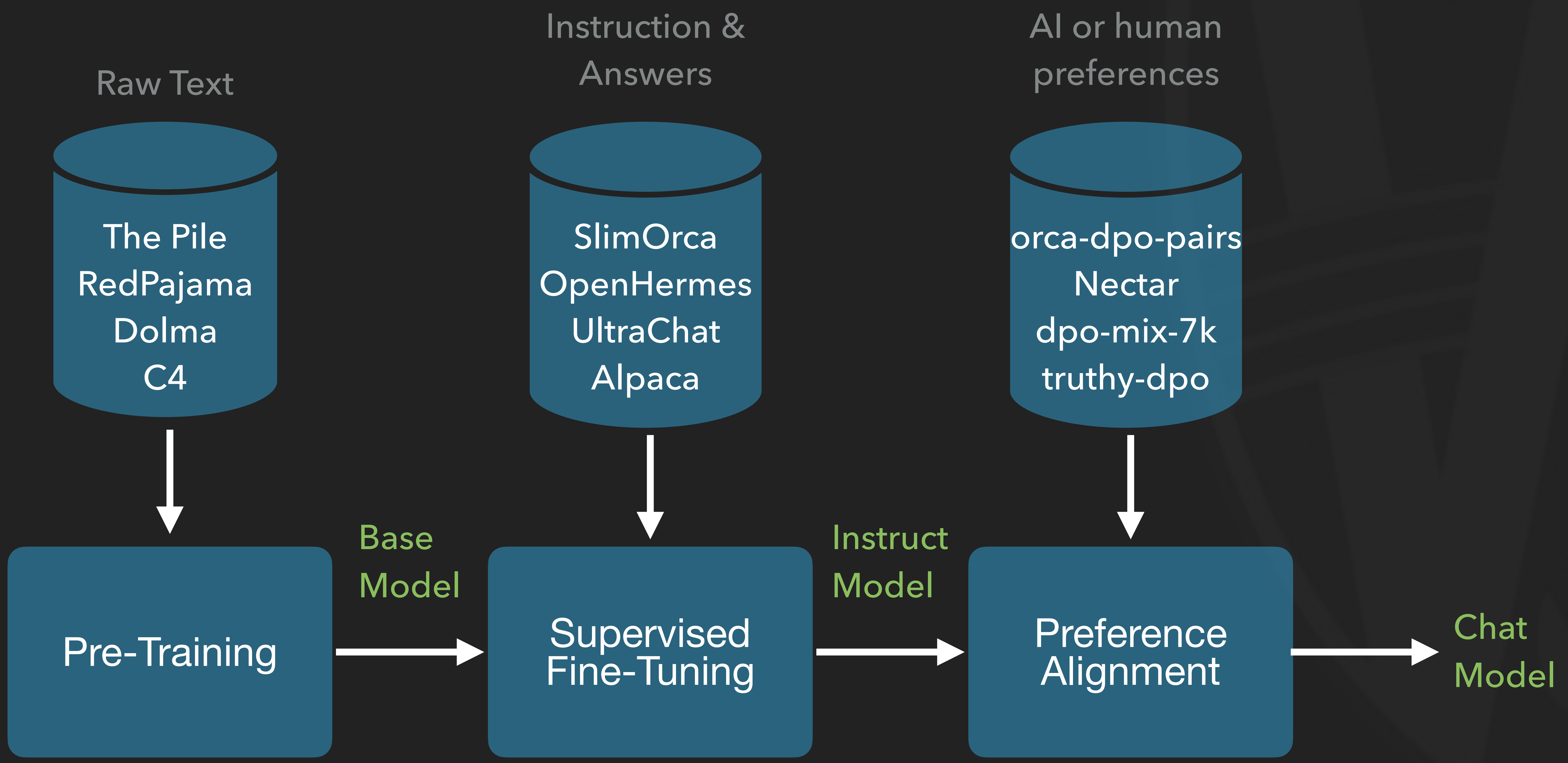
RAG (RETRIEVAL AUGMENTED GENERATION)

- RAG allows us to retrieve snippets of information from external data sources and argument it to user's prompts to get tailored responses from Llama2

- LangChain: A framework that makes it easier to implement RAG



SUPERVISED FINE-TUNING MODELS



THANKS FOR LISTENING

Next : more about LangChain